Lecture 4

What do words mean?

Sequence and Gene Ontology

BMMB 852: Applied Bioinformatics

What analyses are about

In general, most bioinformatics-oriented analyses results fall into two categories:

1. **What a piece of DNA is** - annotation analysis
2. **What a piece of DNA does** - functional analysis

If you do research think about which category does your analysis fall into.

"Ambitious" projects do both - usually ending up with a worse results for each 😱

Famous saying in computing

Premature optimization is the root of all evil

> *The real problem is that programmers have spent far too much time worrying about efficiency in the wrong places and at the wrong times; premature optimization is the root of all evil (or at least most of it) in programming*

Quote by *Donald Knuth* in the *The Art of Computer Programming*.

Adaptation to the current state of biology (my opinion)

Overambitious biology is the root of all evil

> *The real problem is that biologist have spent far too much time worrying about not being exhaustive enough in all the wrong places and at the wrong times; overambition is the root of all evil (or at least most of it) in biology -- adapted from Donald Knuth*

Advice: learn to read between the lines, recognize which statements are based on objective observations and which is wishful thinking.

Why using the correct words matter.

It is difficult to make stuff up when using accurate words!

Ask someone to define "gene".

Now ask someone else.

Most likely it won't be the same definition.

What is a gene?

Are genes concepts or physical entities?

Most scientists use the word "gene" to describe radically different "things".

Biology has many special words

Take the SGD_features.tab file for yeast annotations:

```
wget http://downloads.yeastgenome.org/curation/chromosomal
```

The second column of the file contains the *type* :

```
cat SGD_features.tab | cut -f 2 | sort | uniq
```

produces words like:

```
ORF
CDS
ARS
X_element_combinatorial_repeat
```

What do `ARS`, `CDS` mean?

Is a `CDS` included in an `ORF`?

Ontology

A structured vocabulary that provides:

1. **definitions** - what is a "thing"
2. **classifications** - taxonomy, relationships

Intended to remove ambiguity in the terminology

Important: there may be multiple ontologies describing the same domain of knowledge from different perspectives.

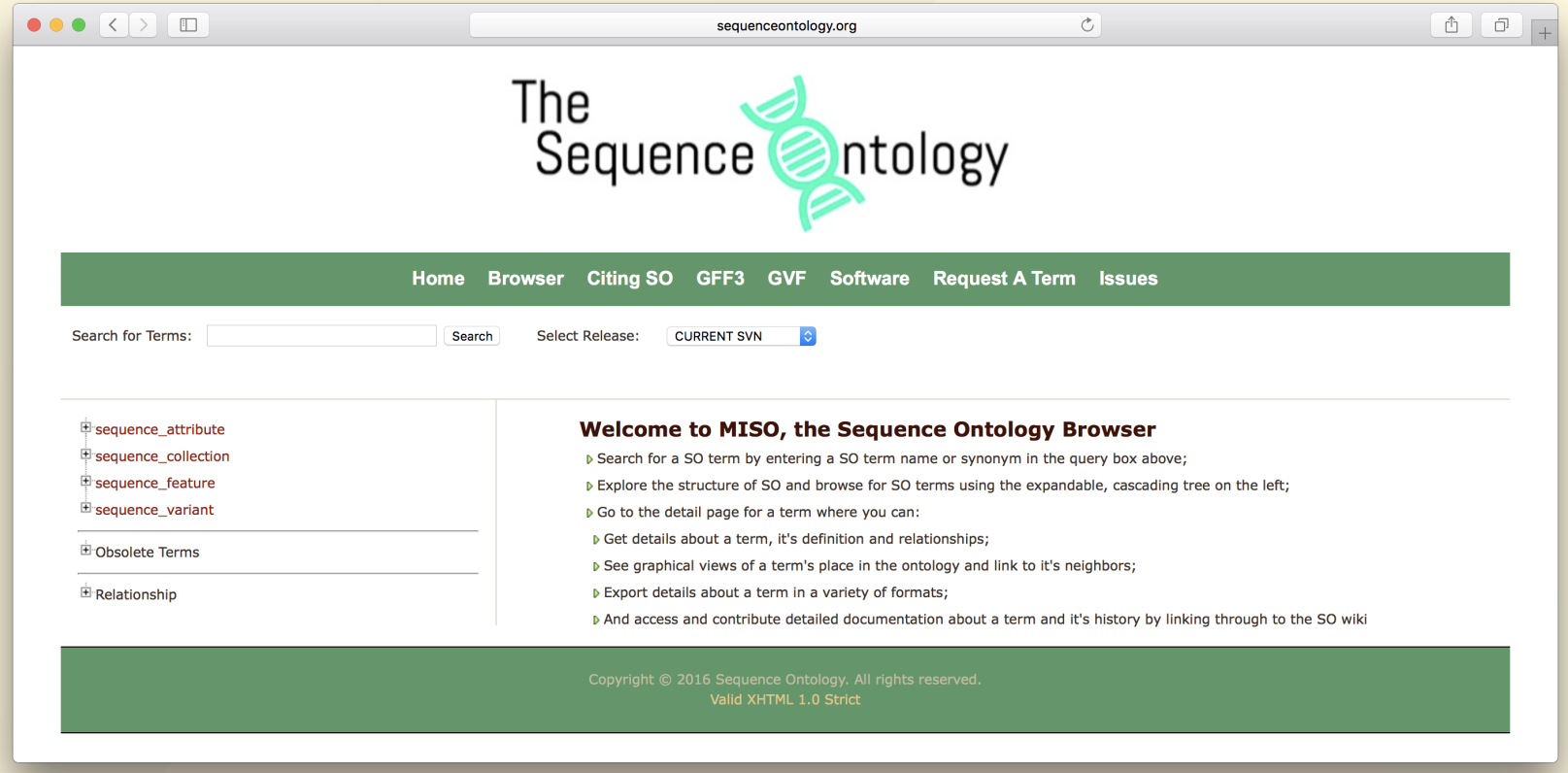
Biological ontologies

In this course we will deal mainly with two types of ontologies:

- **Sequence Ontology (SO)** deals with the definition of biological terms: What is a gene, What is a transcript. Is a transcript part of a gene?
- **Gene Ontology (GO)** deals with the functional characterization of genes. How many different functions are there? Which functions are similar? How do we group functions into categories?

The Sequence Ontology

The Sequence Ontology (SO) defines sequence features used in annotation.



So what is an `X_element_combinatorial_repeat`?

Look it up in the browser

Search the [Sequence Ontology Browser]

X_element_combinatorial_repeat (CURRENT_SVN)	
SO Accession:	SO:0001484 (SOWiki)
Definition:	An X element combinatorial repeat is a repeat region located between the X element and the telomere or adjacent Y' element.
Synonyms:	X element combinatorial repeat
DB Xrefs:	URL: http://www.yeastgenome.org/help/glossary.html
Parents:	repeat_region (SO:0000657)
	telomere (SO:0000624)

> *An X element combinatorial repeat is a repeat region located between the X element and the telomere or adjacent Y' element.*

The definition is built from other terms

The definition may contain other terms that you may not know:

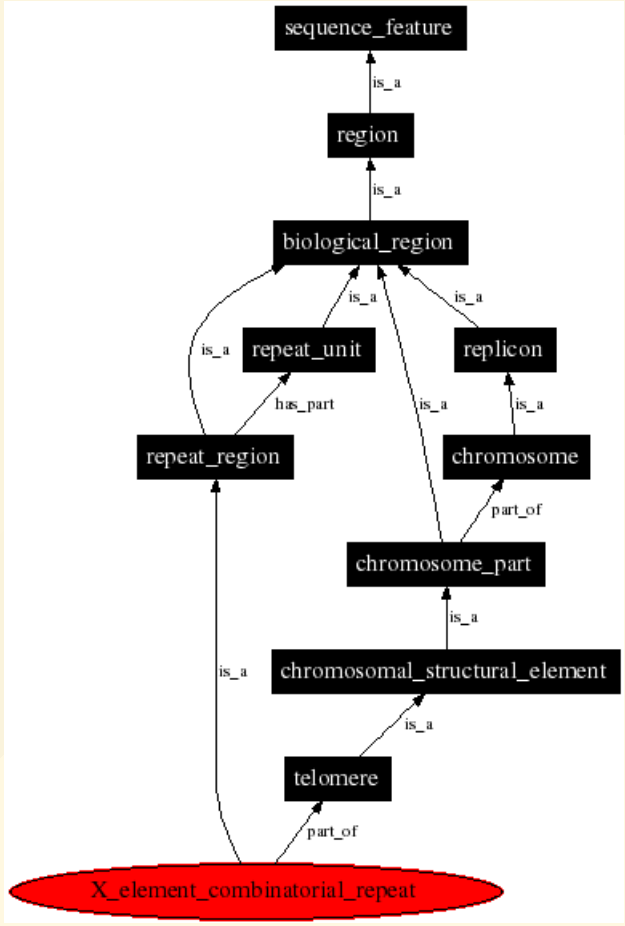
> *An `X element combinatorial repeat` is a `repeat region` located between the `X element` and the `telomere` or adjacent `Y' element`.*

So what is a:

- `repeat unit`, `X element`, `telomere`, `Y element`?

Keep looking up each until you understand what each means.

Graph (tree) representation



The ontology is also a hierarchical classification.

What is a "gene"?

The sequence ontology states:

> *A region (or regions) that includes all of the sequence elements necessary to encode a functional transcript.*
>
> *A gene may include regulatory regions, transcribed regions and/or other functional sequence regions.*

It is much broader concept than what most think.

What is the Gene Ontology (GO)?

The Gene Ontology (GO) is a controlled vocabulary that connects a gene product to one or more functions.

Calling it "Gene Ontology" is misleading.

GO categorizes gene products (proteins) rather than the genes themselves.

Should have been called "Protein ontology"

GO is more complicated than the SO.

It has three, independent categories

How is the GO designed?

The GO project has three independent sub-ontologies:

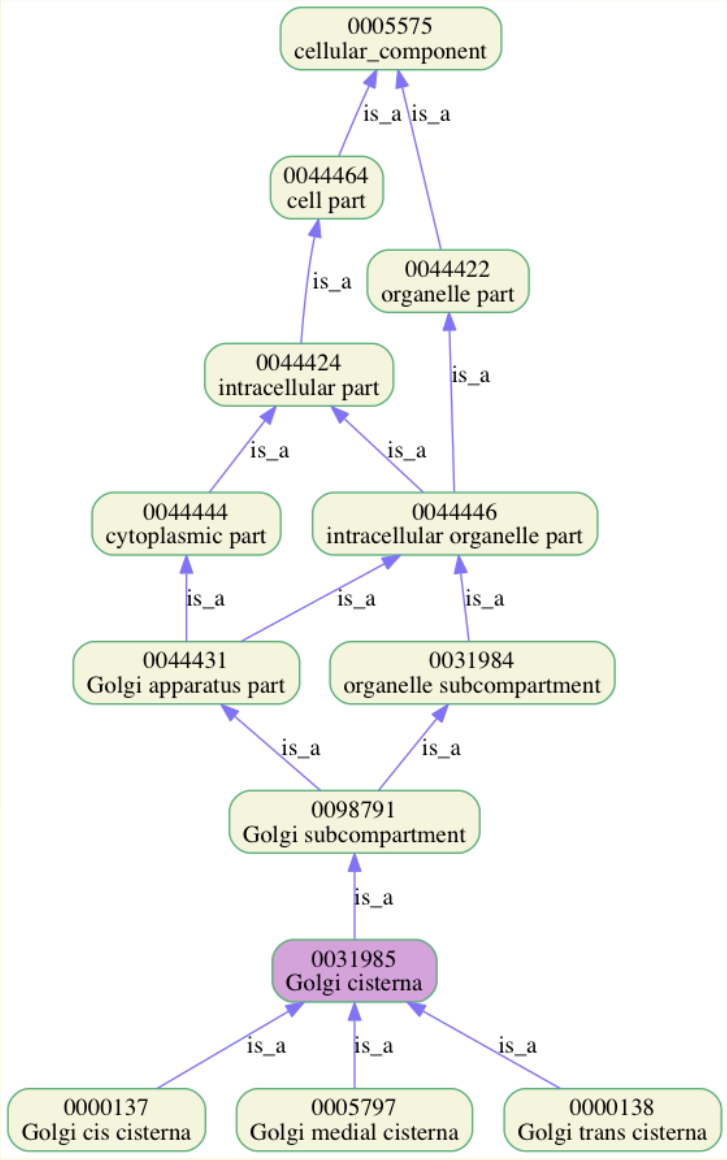
1. **Cellular component (CC).** Where does the product exhibit its effect? -> *cell, nucleus, Golgi membrane*
2. **Molecular function (MF).** How does it work at the molecular level? -> *lactase activity, actin binding*
3. **Biological process (BP).** What is the purpose of the gene product? Involves more than one distinct step: *transport, mitotic prophase, cholesterol efflux*

Where can the Gene Ontology be viewed?

The Gene Ontology website is the authoritative source for definitions, but is not particularly well suited for data interpretation.

The Quick GO service from the European Bioinformatics Institute offers a web interface with more user-friendly functionality.

What do GO relationships look like?



Association files

The first role of GO is to define functions.

The second role is to connect the functions to observed gene products.

The connections are called association files.

A gene product ID is connected to one or more GO functions.

Each organsims will have separate association files.

Gene Ontology (GO) summary

- The GO defines the words used to describe functions.
- The GO also stores the deposited knowledge on different organisms.
- The GO and the associations change over time.

The GO association files represent the accumulated knowledge of life sciences over many decades.

It is among the most essential components of life sciences! Yet most scientists know very little about it - or that it even exists.

Key concepts to remeber

For a typical analysis you need to use both.

First you need concepts from the **Sequence Ontology (SO)** â€“ What types of features are under study? How are the types interrelated?

Then you need concepts from the **Gene Ontology (GO)** â€“ What does a feature do? How does it do it? Where does it do it?

Graphs and networks are best understood when visualized

Use the web interfaces or the local command line tools

to familiarize yourself with their structure